

# Visual Question Answering on VizWiz with a Self-Designed Cross-Modal Attention

DL Basic 2026 Spring — Final Assignment (Competition)

## 1. Overview

We build a Visual Question Answering (VQA) model from scratch for the VizWiz 2023 split (19,873 train / 4,969 test). Given an image and a question, the model predicts an answer string; performance is measured with the standard VQA accuracy  $\text{Acc}(\text{ans}) = \text{mean}_i \min(\#\text{annotators}_{\{j \neq i\}} \text{ that said } \text{ans} / 3, 1)$ .

Our model is a modular, config-driven architecture: a swappable **image encoder** and **text encoder** produce token/region sequences that are combined by a swappable **fusion** module and classified over a closed answer vocabulary. Pretrained backbones are used *only as components* and fine-tuned by our own training loop — no end-to-end VQA model is used. The components are ViT-B/16 (ImageNet-1k pretrained) and BERT-base (Wikipedia + BookCorpus pretrained); **neither was trained on VizWiz or any VQA data** — the answer prediction is learned solely from the distributed VizWiz training set by our own loop. The contribution highlighted here is a **self-designed cross-modal attention fusion**, evaluated against a concat baseline in an ablation study.

## 2. Method

### 2.1 Encoders (unified (features, mask) interface)

- **Image** (swappable): scratch ResNet18 (baseline), ImageNet-pretrained ResNet50, ViT-B/16, or a CLIP ViT-B/16 vision encoder. For CNNs we keep the spatial feature map before global pooling; for ViT/CLIP we keep the patch tokens — both give region tokens  $V \in \mathbb{R}^{B \times N \times d_v}$ . Normalization is read from each backbone's own pretraining config — ImageNet stats for ResNet/ConvNeXt, **(0.5, 0.5, 0.5)** for ViT, CLIP stats for CLIP — rather than assuming ImageNet for all; correcting the ViT statistics gave a measurable test-score gain.
- **Text**: the baseline encodes the question as a multi-hot vocabulary vector projected by a linear layer (one token, order-insensitive). The improved model uses fine-tuned **BERT**, giving contextual token embeddings  $Q \in \mathbb{R}^{B \times L \times d_q}$  with an attention mask.

### 2.2 Cross-modal attention fusion (centerpiece)

Both modalities are projected to a shared width  $H$ :  $V' = V W_v$ ,  $Q' = Q W_q$ . The question tokens **attend over the image regions** with multi-head attention

```
A = MultiHeadAttention(query = Q', key = V', value = V')      # (B, L, H)
Q_ctx = LayerNorm(Q' + A)                                     # residual keeps the question
signal
```

We then pool a masked mean of the attended question tokens and of the image regions and fuse them:

```
z = MLP( [ mean_mask(Q_ctx) ; mean(V') ] )                  # (B, H)
logits = W_cls z
```

Intuition: instead of squashing each modality to one vector and concatenating (which discards *where* in the image the question refers to), each question token pulls the image regions it needs, so the fused representation is grounded in question-relevant image content.

## 2.3 Training

Closed-vocabulary classifier over the full set of normalized training answers (no catch-all `<unk>` class — a catch-all funnels the long answer tail into one dominant, always-wrong class and inflates index-based validation). Loss is cross-entropy on the most-frequent answer (hard label) or the VQA-score soft-label distribution over the 10 annotators (soft label). AdamW (LR 2e-5), cosine schedule with warmup, and AMP. (We also tried differential learning rates — a higher LR for the fresh head, per the course fine-tuning practice — and gradient clipping, but they did not help this model, so the submitted run uses a single LR.) Augmentation is RandomResizedCrop + ColorJitter only — we drop horizontal flip, which can invert the meaning of spatially/text-dependent questions, **and it is applied to training only**; validation uses the deterministic eval transform so val metrics and best-checkpoint selection match real inference. A 10% held-out slice of train gives the VQA-accuracy validation below; the placeholder class is masked at evaluation and inference to mirror the string-matched test.

## 3. Experiments

Dataset, metric, and closed-vocabulary setup as above. We ablate three axes: image encoder (scratch vs pretrained), fusion (concat vs cross-attention), and label (hard vs soft). Validation VQA accuracy is on our held-out train slice; the test column is the Omnicampus score.

### Ablation table

| # | Image encoder         | Text encoder | Fusion          | Label | Extras             | Val VQA acc                          |
|---|-----------------------|--------------|-----------------|-------|--------------------|--------------------------------------|
| 0 | ResNet18 (scratch)    | one-hot      | concat          | hard  | —                  | 0.499 (official baseline, test)      |
| 1 | ResNet50 (pretrained) | one-hot      | concat          | hard  | —                  | 0.5005                               |
| 2 | ResNet50 (pretrained) | BERT         | cross-attention | soft  | —                  | 0.5391                               |
| 3 | ViT-B/16 (pretrained) | BERT         | cross-attention | soft  | ViT norm; 8 epochs | <b>0.5705</b> (test <b>0.57843</b> ) |
| 4 | CLIP ViT-B/16         | BERT         | cross-attention | soft  | diff-LR            | 0.5232                               |

Progression: a pretrained image encoder alone barely helps (0→1, +0.1). Replacing one-hot with BERT and concat with cross-attention (+ soft labels) is the largest jump (1→2, **+3.9 points**); a stronger ViT backbone adds a further +1.4 (2→3). Row 3 is our submitted model: correcting the ViT input normalization to its own (0.5,0.5,0.5) pretraining stats (instead of ImageNet) and training to convergence (8 epochs; validation was still rising at 5) lifted the **Omnicampus test to 0.57843** (held-out validation 0.5705) — well above the 0.499 completion line and approaching the ~0.60 target. (Rows 1–2 report augmented-val numbers; row 3's val is the corrected deterministic val, so it is not directly comparable — the test score is the reliable figure.) A CLIP vision encoder (row 4), despite being vision-language aligned, under-performed ViT at equal epochs — its validation was still rising at epoch 5 (under-converged) rather than clearly better, so ImageNet-pretrained ViT remained our best backbone in the available budget. (Row 0 is the official-baseline Omnicampus test score; rows 1–4 are our held-out validation except where a test score is given.)

### Key finding

Upgrading only the image encoder (row 0→1) barely moved the honest validation metric (0.499 → 0.5005): with a one-hot text encoder the model defaults to the dominant "unanswerable" answer. This localizes the bottleneck to the **text representation and fusion**. Replacing one-hot with fine-tuned BERT and concat with the cross-attention fusion (plus soft labels) then delivers the main gain (row 2, +3.9 points), and a stronger ViT backbone plus correct ViT-input normalization takes the submitted model to test 0.57843 — confirming the diagnosis.

## 4. Discussion

The image-encoder upgrade alone (row 0→1) is nearly a no-op on the honest metric, while adding BERT + cross-attention + soft labels (row 2) yields +3.9 points — the text representation and cross-modal grounding, not the visual backbone, are where VizWiz accuracy is won. The model still over-predicts the majority "unanswerable" class (~74% of predictions; a known VizWiz characteristic and a class-imbalance artifact — outputting the majority answer minimizes loss), which bounds accuracy. We tried the obvious fix — re-weighting the loss by inverse answer frequency — but combined with soft labels it *regressed* (0.5534 → 0.53): steering the model off the safe majority makes it emit confident wrong answers, a net loss under the VQA metric. Two more promising levers remain: (i) an open-vocabulary / generative answer head, since the closed-vocabulary classifier structurally cannot emit answers absent from the training set, and (ii) OCR features fed into the question stream — VizWiz has many "what does this say?" questions where reading in-image text directly should help.

## 5. Conclusion

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A modular from-scratch VQA model with a self-designed cross-modal attention fusion. The ablation isolates where accuracy comes from — text representation + cross-modal fusion, not the visual backbone alone. The submitted ViT-B/16 + BERT + cross-attention + soft-label model reaches **0.57843 VQA accuracy on the Omnicampus test set**, clearing the 0.499 completion line by ~8 points. Pretrained encoders are used only as fine-tuned components inside our own model and training loop. Class re-weighting regressed, so the remaining headroom toward ~0.60 lies in an open-vocabulary generative head and OCR features for in-image text — our identified next steps.